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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2023

END SEMESTER EXAMINATION - ODD 2023
M(CS)101 - Engineering Mathematics - I

TIME ALLOTTED: 3Hours

FULL MARKS: 70

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. (1) CO2 BL2

$$\int_0^1 \int_y^{\sqrt{y}} dx dy =$$

- a) 1/2
- b) 1/6
- c) 2/3
- d) 4/3

2. (1) CO1 BL1

If $A = \begin{pmatrix} 5 & 2 & 3 \\ 1 & 0 & 2 \\ 0 & 4 & 0 \end{pmatrix}$, then the sum of the Eigen values of A is

- a) 3
- b) 5
- c) 6
- d) 10

3. (1) CO2 BL2

$u(x, y) = \frac{x^3 + x^2 y}{\sqrt{x} + \sqrt{y}}$ is a homogeneous function of degree ____

- a) 3/2
- b) 5/2
- c) 1/2
- d) none of these

4. (1) CO3 BL3

The pair of the functions which satisfy the conditions of Cauchy's M.V.T. in $[-1, 1]$ is

- a) $\sqrt[3]{x}, x^2$
- b) $\frac{1}{\sqrt{x}}, \sqrt{x}$
- c) $|x|, x$

d) $\sqrt[3]{x+2}, x^2$

5. If $f_x(a,b)=f_y(a,b)=0$, then (a,b) is (1) CO2 BL2

- a) saddle point
- b) critical point
- c) isolated point
- d) none of these

6. Sum of two homogeneous functions is homogeneous if (1) CO2 BL2

- a) They have same degree of homogeneity
- b) They are both continuous
- c) They are both derivable
- d) none of these

7. The eigen values of the following matrix are (1) CO1 BL1

$$\begin{pmatrix} 1 & 4 \\ 4 & 1 \end{pmatrix}$$

- (a) -5, -3 (b) -5, 3 (c) 5, -3 (d) 5, 3

8. The sum of the series $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$ is (1) CO1 BL1

- a)1 b)0 c)1/2 d) does not exist

9. $f(x,y) = \frac{x^3 + y^3}{\sqrt{x^2 + y^2}}$ is a homogeneous function of degree (1) CO1 BL1

- a) 1 b)0 c)2 d)1/2

10. The rank of the given matrix $A = \begin{pmatrix} 30 & 0 \\ 0 & 1 \end{pmatrix}$ is (1) CO1 BL1

- a. 2 b. 3 c. 30 d. 1

11. If $\lambda^3 - 6\lambda^2 + 9\lambda = 4$ is the characteristic equation of a square matrix A, then A^{-1} is (1) CO3 BL3

- a. $A^2 - 6A + 9I$ b. $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}I$ c. $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}$ d. $A^2 - 6A + 9$

12. If $f(x,y)$ is a homogeneous function of degree 3 then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = kf$. What is the (1) CO1 BL2

value of k?

- a) 3
- b) 2
- c) 0
- d) none of these.

Group B

(Short Answer Type Questions)

(Answer any three of the following) 3x5=15

13. Find maxima and minima of the function $f(x,y)=x^3 + y^3 - 12x - 27y + 20$, also find the saddle points. (5) CO2 BL2
14. Test the convergence of the power series $x + \frac{1}{2} \cdot \frac{x^3}{3} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{x^5}{5} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \frac{x^7}{7} + \dots \infty$ (5) CO3 BL3
15. Verify Cayley-Hamilton theorem for the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$. Hence evaluate A^{-1} . (5) CO3 BL3
16. If $u = \cos^{-1}\left\{\frac{x+y}{\sqrt{x}+\sqrt{y}}\right\}$, then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$ (5) CO4 BL5
17. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = -\frac{3}{(x+y+z)^2}$. (5) CO3 BL3

Group C

(Long Answer Type Questions)

(Answer any three of the following) 3x15=45

18. Do as directed (15)
- a) Use mean value theorem to prove that $\frac{x}{1+x^2} < \tan^{-1} x < x$ when $0 < x < \frac{\pi}{2}$ (5) CO3 BL3
- b) Examine whether Rolle's Theorem is applicable for $y=|x|$ in $[-1, 1]$. (5) CO4 BL4
- c) Test the convergence of the series $\frac{1+2}{2^3} + \frac{1+2+3}{3^3} + \frac{1+2+3+4}{4^3} + \dots$ (5) CO4 BL4
19. Do as directed (15)

- a) Find for what values of β and α the system
 $x_1 + 4x_2 + 2x_3 = 1$
 $2x_1 + 7x_2 + 5x_3 = 2$
 $4x_1 + 8x_2 + \alpha x_3 = \beta$,
 has (i) a unique solution (ii) no solution, (iii) many solutions (6) CO4 BL4

- b) Apply elementary row operations to find A^{-1} of the
 following matrix : $\begin{bmatrix} 1 & 1 & 2 \\ 3 & 1 & 1 \\ 2 & 3 & 1 \end{bmatrix}$ (6) CO3 BL3

- c) Reduce the following matrix $\begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 2 \end{pmatrix}$ to Echelon form
 and hence find its rank. (3) CO2 BL2

20. Do as directed (15)

- a) If $u = u\left(\frac{y-x}{xy}, \frac{z-x}{zx}\right)$ show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$ (5) CO2 BL2

- b) $\iint_R xy(x+y) dx dy$
 R being the region bounded by $y^2=x$ and $y=x$ (5) CO4 BL4

- c) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{-3}{(x+y+z)^2}$ (5) CO2 BL2

- 21a. Verify Cayley Hamilton theorem for the matrix A. Express A^{-1} as a polynomial in A and then compute A^{-1} . (8) CO4 BL3

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 3 & -1 \end{pmatrix}$$

- 21b. If $u = \tan^{-1} \frac{x^3 + y^3}{x - y}$, Find the value of (7) CO4 BL5

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2}$$

- 22a. Check whether the matrix is Diagonalisable and calculate the Diagonal matrix (7) CO4 BL3

$$\begin{pmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

22b. Test the convergence of the series

(8) CO4 BL5

$$1 + \frac{1}{2} \frac{x^2}{4} + \frac{1}{2} \frac{3}{4} \frac{5}{6} \frac{x^4}{8} + \frac{1.3.5.7.9}{2.4.6.8.10} \frac{x^6}{12} + \dots$$

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